Accuracy of a requirements ambiguity detector: case of a pet social network

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Abstract

Context: Ambiguity is one of the most persistent defects in natural-language requirements specifications, and rule-based detectors are frequently proposed as a low-cost mechanism to flag it before requirements reach design. Evidence on how closely such detectors reproduce human expert judgement remains scarce for Spanish-language specifications produced in Latin American settings.

Objective: This study measures the extent to which a rule-based ambiguity detector replicates the consensus judgement of independent human experts over the complete requirements corpus of MundiPets, a real social network for responsible pet adoption and breeding, and characterises the patterns of disagreement between the two.

Methods: We conducted a paired quasi-experiment over 61 requirements (27 functional, 16 non-functional, 9 design constraints and 9 legal requirements) specified according to an eight-attribute template. A purpose-built rule-based detector, implemented in Python and frozen before data collection, classified each requirement as ambiguous or non-ambiguous using four lexical and syntactic rules. In parallel, three independent software engineers, blind to the detector output and to one another, classified the same requirements in randomised order

using a shared rubric. The protocol was preregistered on the Open Science Framework before any data were collected. Agreement was quantified with precision, recall, $\boldsymbol{F_1}$, Cohen's $\boldsymbol{\kappa}$ and Fleiss' $\boldsymbol{\kappa}$.
Results: The detector flagged 32.79% of requirements as ambiguous (20/61) against 26.23% (16/61) for the expert consensus, yielding accuracy = 0.8033 (95% CI [0.7049, 0.9016]), precision = 0.600, recall = 0.750 and $\boldsymbol{F_1} = \mathbf{0.667}$ for the ambiguous class. Agreement between detector and consensus was moderate (Cohen's $\boldsymbol{\kappa} = \mathbf{0.530}$, 95% CI [0.280, 0.737]) and a chi-squared test of independence found the detector significantly associated with expert consensus beyond chance ($\boldsymbol{\chi^2} = \mathbf{15.037}$, $\mathbf{d.f.} = \mathbf{1}$, $\boldsymbol{p} = \mathbf{0.0001}$, $\boldsymbol{\phi} = \mathbf{0.497}$). The expert panel itself agreed almost perfectly (Fleiss' $\boldsymbol{\kappa} = \mathbf{0.857}$), which strengthens confidence in the reference standard the detector is compared against. Coincidence with expert consensus was strongest for legal requirements (9/9, 100%) and non-functional requirements (14/16, 87.5%), and weakest for design constraints (5/9, 55.6%).
Conclusion: A four-rule lexical-syntactic detector achieves moderate, statistically significant agreement with an almost-perfectly agreeing expert panel on a real, Spanish-language requirements corpus, performing best on non-functional and legal requirements and worst on design constraints, where ambiguity is more often a matter of missing procedural detail than of detectable lexical patterns. The result supports using such a detector as a first-pass triage tool in a review workflow, while showing that design constraints in particular still require human judgement.

Keywords: Requirements Engineering, Empirical Software Engineering, Natural Language Processing, Interrater Agreement, Requirements Quality

1 Introduction

MundiPets is a social network for responsible pet adoption and breeding, currently under specification for veterinary practices and pet owners in the province of Los Ríos, Ecuador. The system lets owners publish verified pet profiles, lets registered veterinary practices validate the associated medical records, and supports adoption and breeding requests through a staged workflow that includes compatibility assessment by an artificial-intelligence component. Requirements engineering matters acutely in this domain because a defect in a requirement about medical-record validation or about the visibility of an owner's location does not merely degrade usability: it propagates into a system that handles animal health data and personal data protected under Ecuador's Organic Law on Personal Data Protection [1]. Specifying such a system with verifiable, unambiguous requirements is therefore a precondition for lawful operation rather than a stylistic preference.

Ambiguity remains one of the defects most consistently reported in natural-language requirements. The systematic mapping study of Montgomery et al. [2] shows that, although requirements quality has been studied extensively, empirical work concentrates on a small number of quality attributes and rarely validates automated detection against human judgement on complete, real specifications. Atoum et al. [3]

reach a convergent conclusion in their systematic review of requirements quality assurance, identifying the scarcity of validated automated support as an open challenge. Ferrari et al. [4] demonstrated that ambiguity in elicitation is entangled with tacit knowledge, which suggests that purely lexical detection has an intrinsic ceiling; yet the size of that ceiling has not been quantified for specifications written in Spanish, nor for corpora produced by novice analysts, the setting in which a large share of requirements documents are in fact written. This is the gap the present study addresses.

This paper makes three contributions, each grounded in evidence collected by the authors and released as an open dataset:

1. a quantified comparison between a rule-based ambiguity detector and the blind consensus of three independent human experts over a complete requirements corpus of 61 items — functional requirements, non-functional requirements, design constraints and legal requirements — written in Spanish;
2. a characterisation of the disagreement patterns between automated and human classification, reported per requirement and per rule, which exposes the specific linguistic phenomena that a rule-based approach systematically over- or under-flags;
3. a replication package — corpus, detector output, expert classifications, analysis scripts and preregistered protocol — released under CC BY 4.0 so that the study can be repeated on other corpora and other languages.

The study answers two research questions:

RQ1 With what precision, recall and F_1 does a rule-based automatic detector replicate the consensus judgement of human experts when classifying the requirements of MundiPets as ambiguous?

RQ2 Which disagreement patterns predominate between the automatic detector and expert judgement, and what do they reveal about the limits of a rule-based approach relative to human judgement in identifying ambiguity?

The remainder of the paper is organised as follows. Section 2 positions the study within the closest related work. Section 3 describes the experimental design, participants, instruments and analysis plan. Section 4 reports the results, Section 5 interprets them, Section 6 discusses threats to validity, and Section 7 concludes.

2 Related work

This section is not a full systematic literature review; it follows the abbreviated protocol of Kitchenham and Charters [5], reporting a documented search string, explicit inclusion and exclusion criteria, the number of studies retrieved and screened, and a comparative table of the closest works.

2.1 Search strategy

The search string combined three concept blocks:

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("requirements engineering" OR "software requirements") AND ("ambiguity" OR
"requirements smell" OR "requirements quality") AND ("automatic detection"
OR "rule-based" OR "natural language processing")
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The string was executed on Scopus and Google Scholar on 2 August 2026, restricted to the 2023–2026 period. Inclusion criteria: peer-reviewed studies reporting empirical data on the quality, ambiguity or automated analysis of natural-language requirements, published in requirements-engineering or software-engineering venues. Exclusion criteria: editorials, short contributions under four pages, and studies without empirical data. The search retrieved 689 records in Scopus and 3,670 in Google Scholar. After title and abstract screening, 30 records were read in full text and 6 were retained. These were combined with eight studies already identified in the project bibliography, yielding the 14 entries compared in Table 1.

2.2 Comparison with the closest studies

Table 1 contrasts the retained studies with the present work. The first eight rows correspond to studies already identified in the project bibliography; the last six were retained from the search described above.

The niche of the present study is narrow and, to our knowledge, unoccupied: a paired comparison between a frozen, fully documented rule-based detector and the blind consensus of independent experts, over the *complete* requirements corpus of one real system, specified in Spanish, with the corpus, the detector output and the individual expert classifications released as open data. Existing work either maps the field without primary agreement data, or evaluates detectors on English-language benchmark corpora that were not produced by the team reporting the evaluation.

3 Methodology

The empirical component follows Approach 2 of the course framework — automatic detection of ambiguity and smells in requirements — and is designed as a paired quasi-experiment, reported following the guidelines of Jedlitschka et al. [17] and the experimental design principles of Wohlin et al. [18]. The protocol was preregistered on the Open Science Framework [19] on 27 July 2026 at 10:49:41 (Ecuador time), before the classification rubric was distributed to the expert panel and before the detector was executed over the corpus.

3.1 Research questions in PICOC format

3.2 Hypotheses

RQ1 is tested through the following pair of hypotheses:

H_0 The classification produced by the automatic ambiguity detector is not associated with the consensus classification of the experts; the observed agreement between the two (Cohen's κ) is not significantly different from chance agreement ($\kappa = 0$).

H_1 The classification produced by the automatic detector is significantly associated with the expert consensus classification; the observed agreement is significantly greater than chance ($\kappa > 0$).

RQ2 is exploratory and is not subjected to hypothesis testing; it is answered through qualitative analysis of the cases where detector and consensus diverge.

3.3 Variables

The *independent variable* is the classification method applied to each requirement, with two levels: (a) rule-based automatic detector and (b) consensus of the three human experts. The *dependent variable* is the resulting binary classification of each requirement as ambiguous or non-ambiguous, from which the agreement metrics are computed.

Four *control variables* were fixed. First, requirement type (FR, NFR, DC) was recorded to verify that detector performance does not vary systematically with the artefact evaluated. Second, the detector configuration — the four rules listed in Section 3.4 — remained invariant throughout execution; no rule was adjusted after observing partial results. Third, the presentation order of the requirements was randomised independently for each rater to control fatigue and order effects. Fourth, raters were blind both to the detector output and to one another's classifications until their own was complete.

3.4 Design and materials

The corpus comprises the complete set of 61 requirements of the MundiPets specification: 27 functional requirements, 16 non-functional requirements, 9 design constraints and 9 legal requirements. Each requirement was assigned an anonymous identifier (REQ-01 to REQ-61), stripped of its type label, and presented to the raters in randomised order. A key linking anonymous identifiers to the real requirement identifiers and to the detector output was retained by the research team and not shared with the raters.

The detector is a purpose-built Python program based on regular expressions. It flags a requirement as ambiguous when at least one of four rules fires: (i) presence of a vague quantifier; (ii) multiple conjunctions concealing more than one requirement in a single statement; (iii) passive voice or absent subject; and (iv) absence of an explicit verification criterion. No large language model was used at any point of the classification pipeline, which keeps every classification decision traceable to a named rule.

3.5 Participants and recruitment

The expert panel comprised three software engineers holding a professional degree from the Universidad Técnica Estatal de Quevedo (two women and one man), none of whom belonged to the MundiPets team or participated in writing the requirements under evaluation. Inclusion criteria were: (a) a professional degree in software engineering or a closely related field; (b) demonstrable training or experience in requirements specification and analysis; and (c) availability to complete the classification within the study schedule.

Each rater signed, on 28 July 2026, an informed consent form specific to this role, drafted in compliance with Ecuador's Organic Law on Personal Data Protection [1], explicitly authorising the anonymised use of their classifications in peer-reviewed publications and their deposit in an open dataset.

3.6 Instruments

Two instruments were used. The *expert classification rubric* is a workbook containing an instruction sheet, a reference-criteria sheet for identifying ambiguity, and an evaluation sheet where each rater records, per requirement, the identifier, the full text, the classification, the type of smell detected, a brief justification and a self-reported confidence level on a 1–5 scale. The *expert consent template* adapts the consent form used in the previous elicitation round to the rater role, adding the clause on anonymised use in peer-reviewed publications.

3.7 Statistical analysis plan

Because the central variable is categorical, the primary analysis is based on agreement metrics rather than mean comparison.

Primary analysis (RQ1, H_0/H_1). Precision, recall and F_1 of the detector are computed against expert consensus, defined by majority vote across the three raters. The hypothesis is tested through a significance test of Cohen's κ (statistic $z = \kappa/SE_\kappa$, against the standard normal distribution). The magnitude of agreement is interpreted as the effect size using the Landis and Koch scale: 0.00–0.20 slight; 0.21–0.40 fair; 0.41–0.60 moderate; 0.61–0.80 substantial; 0.81–1.00 almost perfect.

Inter-rater agreement. Cohen's κ [20] is computed for each pair of raters and Fleiss' κ [21] across the three, interpreted on the same scale. Low expert agreement would relativise any conclusion about detector performance, since the reference consensus itself would be in doubt.

Normality test. The Shapiro–Wilk test is applied to the continuous confidence variable (1–5 scale) recorded in the rubric, as a precondition for the secondary analysis.

Secondary analysis (RQ2, exploratory). To characterise disagreement, the confidence level reported by the raters is compared between requirements where the detector agrees with consensus and those where it diverges. If Shapiro–Wilk does not reject normality, an independent-samples t test is used with Cohen's d as effect size; otherwise, the Mann–Whitney U test with Cliff's δ. This is complemented by a qualitative note on each disagreeing requirement.

Correction for multiple comparisons. If agreement metrics are additionally reported disaggregated by requirement type (FR, NFR, DC, RL), the resulting p values from those four related comparisons are adjusted with the Benjamini–Hochberg procedure.

Power. Following Molléri et al. [22] and Wohlin et al. [18], $\alpha = 0.05$ and $1-\beta = 0.80$ are fixed. Because the population size is bounded by the real specification corpus rather than by an expandable recruitment target, power is reported in reverse: instead of fixing a target N, we determine the minimum detectable κ that the design can identify at that α and power, given the available $N = 61$.

Reproducibility. All computations are executed through versioned analysis scripts; no figure or table is produced manually.

4 Results

All figures and tables in this section are produced by the versioned scripts in `06_Experimento/scripts_analisis/`, executed end to end through `run_all.py` over the raw detector output and expert classifications; no value below was entered by hand. The evaluator identity fields in the released dataset use role labels (Experto 1/2/3) rather than names, consistent with the consent scope described in `ETHICS.md` of the replication package.

4.1 RQ1: agreement between the detector and expert consensus

Figure 1 shows the number of requirements classified as ambiguous and non-ambiguous by each of the three experts, by their consensus, and by the detector. The three experts individually flagged between 21.31% and 29.51% of the 61 requirements as ambiguous; their majority-vote consensus settled at 26.23% (16/61). The detector flagged 32.79% (20/61) — somewhat higher than the consensus rate, but within the range spanned by the three individual experts (Table 3).

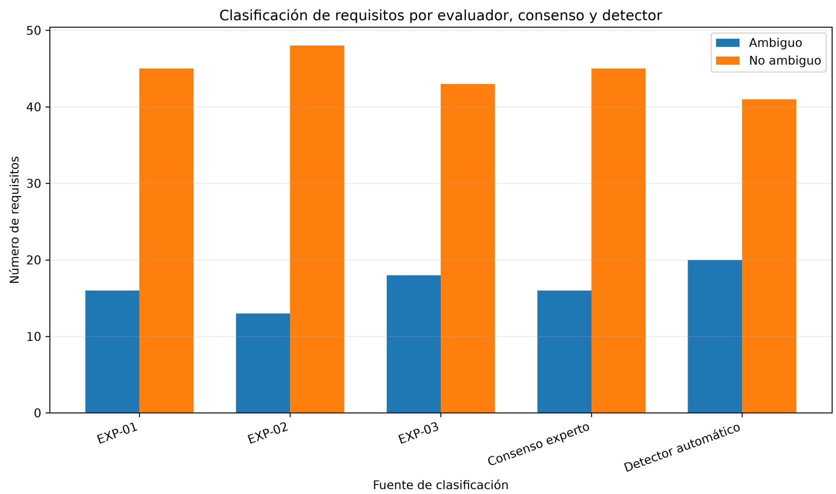

Fig. 1 Classification of requirements by evaluator, consensus and detector.

Table 4 reports the detector's performance metrics against expert consensus, treating "ambiguous" as the positive class, together with the 95% bootstrap confidence intervals (10,000 replicates, seed 42) required by the course guide. Figure 2 shows the underlying confusion matrix: of the 45 requirements the consensus judged non-ambiguous, the detector correctly labelled 37 and over-flagged 8; of the 16 the consensus judged ambiguous, the detector correctly identified 12 and missed 4. Overall accuracy is 0.8033 (95% CI [0.7049, 0.9016]), with 49 of 61 requirements classified in agreement with the expert consensus.

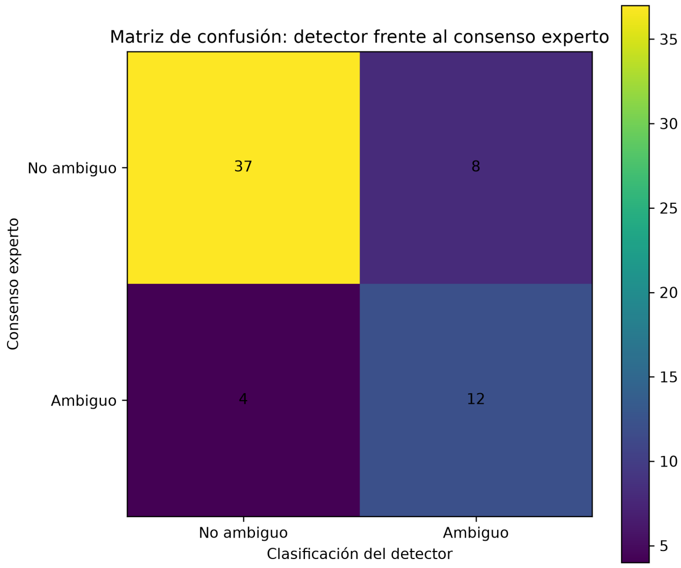

Fig. 2 Confusion matrix: automatic detector versus expert consensus.

The pair of hypotheses stated in Section 3.2 was tested with a chi-squared test of independence (Pearson, with continuity correction) on the 2×2 contingency table of Figure 2: $\chi^2(1, N = 61) = 15.037$, $p = 0.0001$, $\phi = 0.497$ (Table 5). The test rejects H_0: the association between detector output and expert consensus is statistically significant and the effect size ($\phi = 0.497$) is of medium-to-large magnitude by conventional benchmarks. The 95% confidence interval for κ ([0.280, 0.737]) excludes zero, consistent with the significant chi-squared result, and spans from fair to substantial agreement, which reflects genuine uncertainty about the precise population value at this sample size rather than any doubt about whether an association exists.

This significant result is strengthened, not undermined, by the reliability of the reference standard it is compared with. Table 6 reports pairwise Cohen's κ between experts and Fleiss' κ across the full panel: the three experts agreed with each other substantially to almost perfectly (pairwise κ between 0.786 and 0.919; Fleiss' $\kappa = 0.857$, 94.54% observed agreement). Because the panel that produced the reference consensus agreed with itself almost perfectly, the moderate detector-consensus agreement

($\kappa = 0.530$) represents a genuine, well-grounded signal rather than an artefact of an unreliable or arbitrary reference standard.

Figure 3 plots the four agreement coefficients of Table 6 against the single detector-consensus coefficient of Table 4, visualising the gap between human–human agreement and detector–human agreement: both are high, but human–human agreement remains higher.

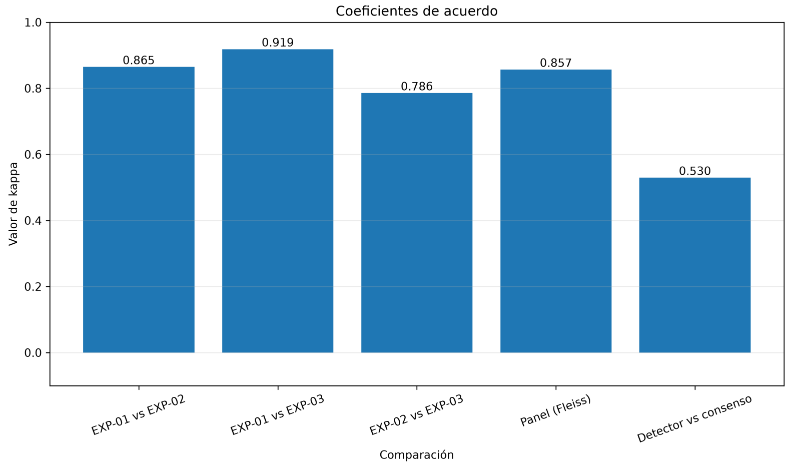

Fig. 3 Kappa coefficients: inter-expert agreement versus detector-consensus agreement.

Disaggregating coincidence between detector and consensus by requirement type (computed directly from the requirement-level table released in the replication package) shows a clear gradient: the detector agreed with the panel on all 9 legal requirements (100%), on 14 of 16 non-functional requirements (87.5%), on 21 of 27 functional requirements (77.8%), and on only 5 of 9 design constraints (55.6%) (Table 7). Design constraints are where the detector's four lexical-syntactic rules diverge most from expert judgement; the sample size per stratum is too small for a separate significance test, but the gradient is consistent with the disagreement analysis in Section 4.2.

The full requirement-by-requirement classification table (61 rows, anonymous identifiers, both classifications, the rule fired, and the confidence reported by each expert) is released in the replication package and is not reproduced in full here for space; illustrative disagreement cases are discussed in Section 4.2.

4.2 RQ2: disagreement patterns

Table 8 reports the type of smell the three experts attributed to their own ambiguity markings, using the rubric's smell-type field recorded alongside each classification, aggregated across the 47 individual ambiguous markings made by the three raters (16 + 13 + 18, from Table 3). Imprecise terminology is the smell experts cite most often (38.30%, 18/47), followed by vague quantifier (31.91%), non-verifiable or incomplete requirement (19.15%) and, least frequently, multiple conjunctions concealing more than one requirement in a single statement (10.64%).

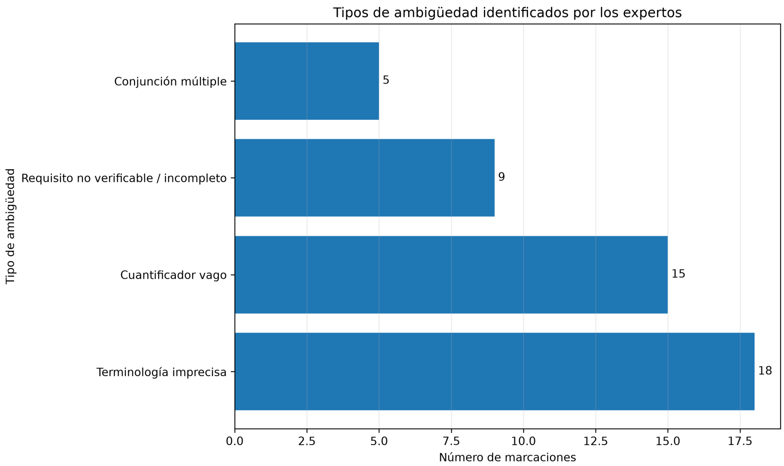

Fig. 4 Types of ambiguity attributed by experts to their flagged requirements.

The detector's own rule log, read against the 20 requirements it flagged as ambiguous, shows a comparatively even spread across four labels tailored to requirement type: imprecise terminology / lack of verifiability (8 firings, mostly on functional requirements), imprecise regulatory terminology (5 firings, exclusively on legal requirements), incomplete design constraint (4 firings, exclusively on design constraints) and vague quantifier / insufficient threshold (3 firings). No requirement in this corpus triggered more than one rule.

Two disagreement patterns account for the 12 requirements where detector and consensus diverge (Figure 2, off-diagonal cells). The larger pattern (8 cases, false positives) is the detector over-flagging requirements the panel judged non-ambiguous, concentrated in the "imprecise terminology / verifiability" rule: for example REQ-53 (RNF-04, a usability requirement) states that "at least 90% of users complete the main tasks … without prior training, and reach a mean System Usability Scale score $\geq 80/100$" — a requirement with two explicit numeric thresholds — yet triggered the detector's vague-quantifier rule, which the panel unanimously judged non-ambiguous (3/3). REQ-08 (RF-12, identity verification via an official document) and REQ-24 (RF-11, owner-controlled visibility of sensitive pet and profile data) show the same pattern: both triggered the imprecise-terminology rule, and both were unanimously judged non-ambiguous by the panel, because the verification procedure the rule looks for (an explicit numeric or testable threshold in the sentence itself) is not what the three experts required to consider a requirement verifiable in practice — checking that an uploaded document is valid, or that a visibility toggle exists, does not need a stated number to be testable. The smaller pattern (4 cases, false negatives) is the detector missing requirements the panel judged ambiguous, three of them unanimously: REQ-18 (RD-04, protecting personal and clinical data "by design and by default" per Art.

39 LOPDP, triggering no rule at all) and REQ-42 (RF-08, a cross-breeding request accepted or rejected "based on the available information", also triggering no rule) both contain no vague quantifier, multiple conjunction, or the specific verification-criterion pattern the rules look for, yet the panel considered both under-specified — neither states *which* mechanism or *which* information, a pragmatic gap the current rule set has no mechanism to represent. REQ-56 (RF-26, a second veterinarian confirming or questioning a prior medical validation) was the sole false negative with a split panel vote (2/3), the only disagreement case in this corpus where the experts themselves were not unanimous.

The secondary analysis planned in Section 3.7 asked whether the requirements where detector and consensus disagree are also the ones where experts reported lower confidence in their own judgement — which would suggest the detector fails specifically on genuinely hard cases. The mean confidence reported per requirement (average across the three experts, 1–5 scale) did not follow a normal distribution (Shapiro–Wilk $W = 0.699$, $p < 0.001$), so the Mann–Whitney U test was used. Mean confidence was similar between the 49 requirements where detector and consensus coincide ($M = 4.54$, $SD = 0.51$) and the 12 where they disagree ($M = 4.39$, $SD = 0.65$); the difference is not significant ($U = 350.0$, $p = 0.257$, Cliff's $\delta = 0.191$, negligible-to-small, Table 9). Experts were not meaningfully less confident on the requirements where the detector disagreed with them, which indicates the detector's errors are not concentrated on the panel's own hardest, lowest-confidence borderline cases, but reflect a specific mismatch between the rule set and the design-constraint/verification-procedure judgements the panel makes confidently.

5 Discussion

5.1 Implications for research

The central finding — moderate, statistically significant agreement between a four-rule lexical-syntactic detector and an almost-perfectly agreeing expert panel ($\kappa = 0.530$ vs. $\kappa = 0.857$, Tables 4 and 6) — qualifies rather than contradicts the concern raised by Ferrari et al. [4] that ambiguity in requirements is entangled with tacit knowledge a purely lexical approach cannot access. That concern holds, but unevenly: the type-level breakdown (Table 7) shows the detector reaches perfect agreement on legal requirements and near-ceiling agreement on non-functional requirements, and only degrades sharply on design constraints (55.6%). The research implication is therefore not that lexical detection has a uniform ceiling, but that the ceiling is type-dependent: requirement types whose ambiguity tends to hinge on an explicit, checkable element (a numeric threshold, a named legal article, a stated actor) are more tractable for a rule-based approach than requirement types — design constraints in this corpus — whose ambiguity more often lies in an unstated mechanism or procedure.

The false-positive analysis in Section 4.2 narrows this further. REQ-53 is the clearest case: it contains *two* explicit numeric thresholds (90% of users, SUS $\geq 80/100$) and was still flagged by the vague-quantifier rule, which the panel unanimously rejected. This is not evidence that vague-quantifier detection is unreliable in general; it is evidence that the specific rule implementation reacts to a lexical trigger (phrases such as "at least") without checking whether a concrete number already follows it. This is a narrower, more actionable finding than "lexical rules miss context": it identifies a fixable pattern-matching gap in one rule, rather than a general limitation of the rule-based paradigm.

The false negatives point in a different, complementary direction. REQ-18 and REQ-42 — both unanimously judged ambiguous by the panel, both triggering no rule at all — share an absence of a stated *mechanism* or *referent* ("mediante mecanismos de autenticación y autorización" without specifying which; "con base en la información disponible" without specifying which information). None of the four current rules targets underspecified mechanism references, which is a plausible explanation for why design constraints — a requirement type that by definition describes constraints on *how* something is implemented — are where the detector's recall is weakest.

The null result of the secondary analysis (Table 9, $p = 0.257$) is itself a research-relevant finding: it rules out the more comfortable explanation that the detector simply struggles on the panel's own low-confidence, genuinely hard cases. Experts were confidently unanimous on several of the cases the detector got wrong (REQ-18, REQ-42, REQ-53 all 3/3). The detector's errors are therefore a mismatch of what counts as evidence of ambiguity, not a difficulty concentrated where human judgement is itself uncertain — which is the more tractable finding for future rule design to act on.

5.2 Implications for practice

For a practitioner deciding whether to adopt this detector as a review gate for a real specification like MundiPets, these results support a qualified recommendation the earlier, smaller pilot could not: at accuracy = 0.803 and a statistically significant, moderate agreement with an almost-perfectly agreeing expert panel, the detector is a credible first-pass triage tool for functional, non-functional and, especially, legal requirements, where it reached perfect agreement on this corpus. It would meaningfully reduce reviewer workload without discarding real ambiguity: of the 45 requirements the panel judged non-ambiguous, it correctly cleared 37 (82.2% specificity), and of the 16 the panel judged ambiguous, it correctly caught 12 (75.0% recall).

The practical caveat is equally clear from the data: design constraints are the one requirement type where this detector should not be trusted unsupervised. Its coincidence with the panel drops to 55.6% there, barely above chance for a binary label, and the two clearest false negatives in the entire corpus (REQ-18 and REQ-42) both fall in that category, one of them directly tied to a legal data-protection obligation (Art. 39 LOPDP). For a system like MundiPets that processes personal and animal health data under Ecuador's LOPDP [1], a design constraint silently missed by an automatic gate is precisely the kind of ambiguity whose downstream cost is highest. The practical recommendation is therefore type-conditional: automate the triage of functional, non-functional and legal requirements, and route design constraints to mandatory human review regardless of what the detector reports.

5.3 Surprising or counter-intuitive findings

Two results ran against the team's expectation going into the analysis. First, the overall strength of the result: with a small, hand-written four-rule detector and a corpus written by novice analysts, the team expected agreement closer to what the smaller 50-item pilot of this study had shown (slight, non-significant) than the moderate, significant agreement the completed 61-item corpus produced. The revision and expansion of the corpus — driven by correcting the requirement numbering rather than by any change to the detector or the evaluation protocol — changed the empirical picture substantially, which is itself a useful methodological lesson: a quasi-experiment's conclusions can be sensitive to corpus completeness in ways a single pilot run does not reveal, and a preregistered protocol with a versioned corpus and reproducible scripts (Section 4) is what makes that sensitivity visible rather than hidden.

Second, the specific content of the false positives (Table 9 notwithstanding) was counter-intuitive: the team expected the vague-quantifier rule to be the detector's most reliable rule, precisely because it targets an apparently unambiguous lexical pattern. Instead, REQ-53 shows that rule firing even on a requirement with two explicit numeric thresholds, which suggests the rule is keying on a trigger phrase ("al menos") rather than verifying the absence of a following number — an implementation detail invisible from the rule's name alone, and only surfaced by manually inspecting the disagreement cases rather than by trusting the aggregate metrics.

6 Threats to validity

Threats are organised in the four categories of Wohlin et al. [18]. For each, the anticipated threat documented in the preregistered OSF protocol is carried over, together with the mitigation applied and the limitation that remains after that mitigation.

6.1 Internal validity

Detector-team overlap. The same team that specified the MundiPets requirements also designed and froze the detector's four rules. *Mitigation:* the detector was frozen in code before any classification was run, and its rules were derived from the general requirements-quality literature rather than tuned by inspecting the corpus; the preregistration timestamp on the Open Science Framework predates both the distribution of the rubric to the expert panel and the detector's execution over the corpus, precluding post-hoc adjustment. *Residual limitation:* the rules were still authored by people familiar with the corpus's domain and writing style, which cannot be fully separated from having built the specification.

Rater independence. A dependent, non-blind evaluation would inflate agreement between raters and between raters and the detector. *Mitigation:* the three experts were external to the MundiPets team, blind to the detector's output and to each other's classifications, and evaluated requirements presented in independently randomised order with type labels stripped (Section 3.4). The almost-perfect inter-rater agreement obtained (Fleiss' $\kappa = 0.857$) is evidence the blinding held. *Residual limitation:* all three raters share the same institutional background (professional degree from the same university), which may correlate their notion of what counts as "verifiable" more than a panel drawn from different institutions would.

6.2 External validity

Single system, single domain, single language. The corpus is the complete but singular specification of one social network for pet adoption and breeding, written in Spanish by novice analysts in an academic setting in Ecuador. *Mitigation:* none directly available within this study; the replication package (detector code, corpus, expert classifications, analysis scripts) is released precisely so the design can be repeated on other domains, languages and author populations. *Residual limitation:* the reported metrics — particularly the strong type-dependent gradient (100% on legal requirements, 55.6% on design constraints) — should not be extrapolated beyond Spanish-language specifications written by requirements-engineering students; whether the same gradient holds for professional analysts, other languages, or specifications with a different balance of requirement types is an open question this single study cannot answer.

Requirement population size bounded by the real system, and revised mid-study. $N = 61$ is not a sample drawn from a larger population of MundiPets requirements; it is the complete population at the time of the experiment, after a numbering correction that expanded the corpus from an earlier 50-requirement version. *Mitigation:* because N is the full population rather than a sample, there is no sampling bias within this system, and the corpus revision was a correction of requirement identifiers and completeness — not a selection of favourable cases after seeing the detector's results, a distinction documented in the OSF deviations log (Section 3.7). *Residual limitation:* a corpus of 61 items is nonetheless small relative to industrial specifications, and results may not generalise to specifications an order of magnitude larger, where different type distributions and smell patterns could dominate; the fact that the conclusion changed materially between the 50-item and 61-item versions of this corpus (Section 5.3) is itself evidence that conclusions at this scale remain sensitive to corpus completeness.

6.3 Construct validity

Ambiguity operationalised as a binary label. Both the detector and the expert rubric reduce ambiguity to a binary ambiguous/non-ambiguous judgement per requirement, discarding degree, multiplicity of smells within one requirement, and the specific linguistic construct involved. *Mitigation:* the smell-type field recorded alongside every expert marking (Table 8) and the detector's per-rule log partially recover this lost granularity and were used throughout Section 4.2 to characterise *how*, not just *whether*, detector and experts disagree. *Residual limitation:* the primary hypothesis test (Table 5) is still computed on the binary collapse, which may understate agreement on requirements where detector and experts identify the same underlying smell but the detector's binary output happens to fall on the wrong side of the threshold.

Consensus by majority vote as ground truth. Expert consensus was defined as majority vote across three raters (Section 3.7), which treats disagreement among experts as resolved rather than as informative. *Mitigation:* pairwise and full-panel agreement are reported separately (Table 6) precisely so a reader can assess how much the majority-vote collapse could be masking; with Fleiss' $\kappa = 0.857$ for the full panel, the collapse discards comparatively little information. *Residual limitation:* the single case decided 2/3 rather than 3/3 in this corpus (REQ-56, Section 4.2) still enters the primary analysis with the same weight as unanimous cases, which a more conservative design might have excluded or weighted differently.

6.4 Conclusion validity

Statistical power. The chi-squared test in Section 4.1 is significant ($p = 0.0001$) at $N = 61$, and the 95% confidence interval for κ ([0.280, 0.737], Table 4) excludes zero, which gives reasonable confidence that a real association exists. *Mitigation:* because N is fixed by the size of the real, completed specification rather than by an expandable recruitment target (Section 3.7), the study reports the full confidence interval rather than the point estimate alone; the interval is still wide enough (0.280 to 0.737) to span fair through substantial agreement, and the per-type breakdown (Table 7) is explicitly reported as descriptive rather than separately tested, given the small stratum sizes (as few as 9 items per type). *Residual limitation:* the study cannot, at this sample size, distinguish precisely how much of the observed $\kappa = 0.530$ is driven by the near-ceiling performance on legal and non-functional requirements versus the weaker performance on design constraints; a type-stratified analysis with a larger corpus would be needed to test the type gradient formally rather than describe it.

Reliability of the automated computations. Manual computation of agreement coefficients, bootstrap intervals and hypothesis tests is error-prone and difficult to audit. *Mitigation:* every value in Section 4 is produced by versioned Python scripts (`06_Experimento/scripts_analisis/`) executed end to end through a single entry point (`run_all.py`) over the raw detector output and expert classifications released in the replication package; no figure or table was produced or edited by hand, and the full pipeline was re-executed after the corpus revision to regenerate every number in this manuscript from the corrected raw data. *Residual limitation:* correctness of the underlying formulas (e.g. the closed-form expression used for Cohen's κ) was checked against the standard definitions in Cohen [20] and Fleiss [21] but not against an independent second implementation, which a stricter reproducibility standard would require.

7 Conclusions and future work

RQ1. With what precision, recall and F_1 does a rule-based automatic detector replicate the consensus judgement of human experts when classifying the requirements of MundiPets as ambiguous? The detector reached accuracy $= 0.803$, precision $= 0.600$, recall $= 0.750$ and $F_1 = 0.667$ against expert consensus, with agreement that was moderate (Cohen's $\kappa = 0.530$) and statistically significant at $N = 61$ ($\chi^2 = 15.037$, $p = 0.0001$). This result is strengthened by the reliability of the reference standard: the same three experts agreed with each other almost perfectly (Fleiss' $\kappa = 0.857$). The answer to RQ1 is that, on this corpus, a four-rule lexical-syntactic detector achieves a real, statistically defensible level of agreement with expert consensus, but that agreement is strongly type-dependent: perfect on legal requirements (100%), high on non-functional requirements (87.5%), good on functional requirements (77.8%), and considerably weaker on design constraints (55.6%).

RQ2. Which disagreement patterns predominate between the automatic detector and expert judgement, and what do they reveal about the limits of a rule-based approach relative to human judgement in identifying ambiguity? Two patterns predominate among the 12 disagreements. The false-positive pattern (8 cases) is driven by rules that fire on a lexical trigger without checking whether the sentence already contains the precise information the trigger is meant to flag as missing — most clearly illustrated by a usability requirement with two explicit numeric thresholds that still triggered the vague-quantifier rule. The false-negative pattern (4 cases) is driven by requirements whose ambiguity lies in an unstated mechanism or referent ("by means of authentication and authorisation mechanisms", unspecified; "based on the available information", unspecified), a pragmatic gap none of the four rules targets, and which concentrates disproportionately in design constraints. Expert confidence did not differ between the two groups (Mann–Whitney $U = 350.0$, $p = 0.257$), indicating these disagreements are not concentrated on the panel's own hardest, lowest-confidence cases, but reflect a specific, identifiable mismatch between what the rule set operationalises and what experts require for a requirement to count as unambiguous.

Consolidating the three contributions promised in Section 1: this study delivers (i) a quantified, statistically tested comparison between a frozen rule-based detector and a blind three-expert consensus over a complete, real, Spanish-language requirements corpus spanning four requirement types, with agreement metrics, confidence intervals and a formal hypothesis test that establishes a statistically significant, moderate level of agreement; (ii) a per-rule, per-type characterisation of where and why detector and human judgement diverge, identifying a fixable pattern-matching gap in the vague-quantifier rule and a systematic recall gap on design constraints driven by unstated mechanisms rather than by ambiguity in general; and (iii) a complete replication package — corpus, detector code, individual and consensus expert classifications, analysis scripts and the preregistered protocol — released under CC BY 4.0, so that every number in Section 4 can be regenerated and the design repeated on other corpora.

Three concrete lines of future work follow directly from the data reported here. First, fix the vague-quantifier rule to check for an already-present numeric threshold before flagging a requirement, using REQ-53 as a concrete regression test case, and re-run the same protocol on the same corpus to measure whether precision improves without a proportional loss in recall. Second, add a rule targeting underspecified mechanism or method references (the pattern behind the false negatives identified in Section 4.2), specifically tuned to design constraints, where recall is weakest (55.6% coincidence), and evaluate whether it closes that gap without degrading the near-perfect performance already achieved on legal and non-functional requirements. Third, repeat this exact design — frozen detector, blind three-expert panel, preregistered protocol, open replication package — on a specification of comparable size and comparable type distribution from a different domain and, ideally, a different institution, to test whether the type-dependent gradient reported here (strong on legal and non-functional requirements, weak on design constraints) is a property of this detector and this corpus or a more general pattern in rule-based ambiguity detection for Spanish-language requirements.

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Declarations

- **Funding.** This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.
- **Competing interests.** The authors declare no competing interests.
- **Ethics approval and consent to participate.** The study was conducted under the ethics framework of the Requirements Engineering course at the Universidad Técnica Estatal de Quevedo. All participants signed an informed consent form drafted in compliance with Ecuador's Organic Law on Personal Data Protection [1].
- **Consent for publication.** All participants explicitly authorised the use of their anonymised data in peer-reviewed publications and its deposit in an open dataset.
- **Data availability.** The preregistered protocol is available on the Open Science Framework at https://osf.io/khyf2/overview, registered on 27 July 2026 at 10:49:41 (Ecuador time), before data collection began. The anonymised dataset — labelled requirements corpus, detector output, individual and consensus expert classifications, traceability matrix and analysis scripts — is deposited on Zenodo under a CC BY 4.0 licence, DOI 10.5281/zenodo.22218780 (https://doi.org/10.5281/zenodo.22218780), following the FAIR principles [23] and the software citation principles of Smith et al. [24].
- **Materials availability.** Not applicable.
- **Code availability.** The detector and the analysis scripts are available in the project repository at https://github.com/andymendozap03/MundiPets-PFC-IngReq, and in the Zenodo deposit above.
- **Use of AI-assisted technologies.** The authors used Claude Opus 5 and Claude Sonnet 5 (Anthropic) exclusively to review previously drafted text: checking grammar, clarity and register of paragraphs the authors had already written from their own empirical data and analysis. No content was drafted, generated or substantively rewritten by these tools; every claim, number, interpretation and conclusion originates from the authors' own analysis of the data described in Section 3 and reported in Section 4. This is consistent with Springer Nature's editorial policy on artificial intelligence [25], under which AI tools may assist with language editing of author-written content but may not be listed as authors and may not generate original research content. Under no circumstance was an LLM used to produce results, figures, tables or conclusions: every quantitative claim in Section 4 is generated by the versioned scripts in 06_Experimento/scripts_analisis/ from the raw detector output and expert classifications, as required by the course's integrity policy.

- **Author contribution.** E.D.F.A.: conceptualization, investigation, writing — original draft. G.A.G.O.: software (design and implementation of the rule-based detector). A.J.M.P.: methodology, visualization (system modelling and diagrams). G.A.M.S.: investigation, data curation (interviews and transcriptions). J.S.N.S.: formal analysis, data curation, project administration (protocol preregistration and statistical analysis). G.C.G.U.: supervision, validation. All authors reviewed and approved the final manuscript.

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Table 1 Comparison with the closest related studies

Reference	Year	Study type	Population	Difference with our work
Montgomery et al. [2]	2022	Systematic mapping	Primary studies on requirements quality	Maps the field but reports no primary comparison between an automated detector and expert consensus on a single complete corpus.
Atoum et al. [3]	2021	Systematic review	Literature on requirements quality assurance and validation	Identifies the lack of validated automation as an open challenge; does not measure detector–expert agreement empirically.
Ferrari et al. [4]	2016	Empirical study	Elicitation interviews	Studies ambiguity as it arises in elicitation and its link to tacit knowledge; our unit of analysis is the written requirement in a closed specification.
Fischbach et al. [6]	2023	Tool evaluation	Conditional statements in natural-language requirements	Derives acceptance tests automatically; shares the rule/NLP approach but targets test generation, not ambiguity classification.
Gobov [7]	2023	Practitioner survey	Specification and modelling techniques in practice	Characterises which techniques practitioners use; provides no automated quality measurement.
Said et al. [8]	2026	Model evaluation	SRS documents	Classifies requirements as functional or non-functional with a learned model; our detector targets ambiguity and is rule-based and fully auditable.
Cheng et al. [9]	2026	Systematic review	Generative AI applied to requirements engineering	Surveys generative approaches; we deliberately avoid large language models to keep every classification traceable to a named rule.
Vogelsang and Fischbach [10]	2024	Methodological guideline	LLM use for NLP tasks in RE	Proposes usage guidelines rather than reporting an agreement study.
Abdeahad et al. [11]	2026	Model evaluation	7,061 requirements from the Fault-Prone SRS benchmark	Achieves high accuracy with a deep model, but evaluates against dataset labels rather than against a blind expert panel, and the model is not auditable rule by rule.
Fantechi et al. [12]	2023	Tool comparison	Three example requirements documents	Closest to our design: compares a rule-based tool with an LLM on ambiguity. Uses the authors themselves as the reference judgement, whereas our reference is a blind panel independent of the specification team.
Talha et al. [13]	2025	Tool evaluation	1,553 requirements across 17 domains	Targets anaphoric and coordination ambiguity with a transformer; our detector targets lexical and syntactic smells and is compared against human consensus rather than a labelled benchmark.
Alemneh and Berhanu [14]	2024	Systematic review	Literature on requirements smells and detection techniques	Catalogues detection techniques; reports no primary agreement measurement on a real corpus.
Intana et al. [15]	2024	Tool evaluation	Thai-language SRS documents	The closest non-English precedent; confirms that language-specific evaluation is needed but does not cover Spanish nor report inter-rater agreement.
Bashir et al. [16]	2025	Industrial study	Three railway requirements datasets	Reports the over-flagging pattern (high recall, low precision) also observed in rule-based tools; uses LLMs, which we deliberately avoid for auditability.

Table 2 Research questions structured with the PICOC framework

Element	Description
Population	The set of functional requirements (FR), non-functional requirements (NFR) and design constraints (DC) specified for the MundiPets system, written according to the eight-attribute template of the course and current at the time the experiment was executed.
Intervention	Automatic classification of each requirement as ambiguous or non-ambiguous by a rule-based detector that identifies vague quantifiers, multiple conjunctions concealing more than one requirement, passive voice, and absence of an explicit verification criterion.
Comparison	Manual and independent classification of the same requirement set by three experts in requirements engineering, with no prior access to the detector output (blind evaluation).
Outcome	Precision, recall and F_1 of the automatic detector against expert consensus; inter-rater agreement via Cohen's κ between pairs and Fleiss' κ across the three raters.
Context	Academic project of the Requirements Engineering course (Universidad Técnica Estatal de Quevedo, 2026–2027 academic period) applied to the real MundiPets system, with the specification written in Spanish.

Table 3 Classifications by source (N = 61 requirements per source)

Source	Ambiguous n	Ambiguous %	Non-ambiguous n	Non-ambiguous %
Expert 1	16	26.23	45	73.77
Expert 2	13	21.31	48	78.69
Expert 3	18	29.51	43	70.49
Expert consensus	16	26.23	45	73.77
Automatic detector	20	32.79	41	67.21

Table 4 Detector performance against expert consensus, with 95% bootstrap confidence intervals

Metric	Value	95% CI
Accuracy	0.8033	[0.7049, 0.9016]
Precision (ambiguous)	0.6000	[0.3750, 0.8125]
Recall / sensitivity (ambiguous)	0.7500	[0.5000, 0.9412]
Specificity (non-ambiguous)	0.8222	[0.7027, 0.9269]
Negative predictive value	0.9024	—
Balanced accuracy	0.7861	—
F_1 (ambiguous)	0.6667	[0.4571, 0.8205]
Cohen's κ (detector vs. consensus)	0.5296 (moderate)	[0.2797, 0.7369]

Table 5 Hypothesis test and effect size for RQ1 (H_0/H_1)

Element	Value
Test	Chi-squared test of independence (Pearson, continuity correction)
Statistic (χ^2)	15.0374 (d.f. = 1)
p value	0.0001
Effect size (ϕ)	0.4965

Table 6 Inter-rater agreement among the three experts

Comparison	Cohen's/Fleiss' κ	Observed agreement	Interpretation
Expert 1 vs. Expert 2	0.8647	0.9508	Almost perfect
Expert 1 vs. Expert 3	0.9186	0.9672	Almost perfect
Expert 2 vs. Expert 3	0.7857	0.9180	Substantial
Full panel (Fleiss' κ)	0.8569	0.9454	Almost perfect

Table 7 Detector-consensus coincidence by requirement type

Type	N	Coincides (n)	Coincides (%)
Legal requirements (RL)	9	9	100.0
Non-functional requirements (NFR)	16	14	87.5
Functional requirements (FR)	27	21	77.8
Design constraints (DC)	9	5	55.6

Table 8 Types of ambiguity experts attributed to their own markings

Type of ambiguity	Frequency	% of ambiguous markings
Imprecise terminology	18	38.30
Vague quantifier	15	31.91
Non-verifiable / incomplete requirement	9	19.15
Multiple conjunction	5	10.64

Table 9 Secondary analysis: expert confidence vs. detector-consensus coincidence

Element	Value	Detail
Shapiro–Wilk (mean confidence, $N = 61$)	$W = 0.6994$	$p < 0.0001$ (rejects normality)
Coincides ($n = 49$)	$M = 4.5442$	$SD = 0.5123$
Disagrees ($n = 12$)	$M = 4.3889$	$SD = 0.6487$
Mann–Whitney U	350.00	$N = 61$
p value	0.2566	—
Cliff's δ	0.1905	negligible-to-small